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D15C 16
MLDL

Experiment No 02

Aim: Implement Multi Regression, Lasso, and Ridge Regression on real-world datasets

1. Dataset Source

The dataset used for this experiment is the **Medical Insurance Cost Dataset**, which is publicly available on Kaggle.

Dataset Link:
https://www.kaggle.com/datasets/rahulvyasm/medical-insurance-cost-prediction

This dataset is commonly used to study how personal and lifestyle factors affect medical insurance charges and is well-suited for regression-based prediction tasks.

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2. Dataset Description

The Medical Insurance dataset contains information about individuals and the corresponding medical insurance charges they incur. The objective of this experiment is to predict the **insurance charges** based on personal and health-related attributes.

Dataset Characteristics

- Number of instances: 1338 records
- Type: Structured dataset
- Domain: Healthcare / Insurance Analytics

Features Used

- **Age** – Age of the individual
- **Sex** – Gender of the individual (male/female)
- **BMI** – Body Mass Index, an indicator of body fat
- **Children** – Number of children covered by insurance
- **Smoker** – Smoking status (yes/no)
- **Region** – Residential area in the US

Categorical features such as *sex*, *smoker*, and *region* are converted into numerical form using one-hot encoding before training the models.

Target Variable

- **Charges** – Medical insurance cost billed to the individual

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3. Mathematical Formulation of the Algorithms

Multiple Linear Regression

Multiple Linear Regression models the relationship between a dependent variable and multiple independent variables.

Equation:

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_{\square} x_{\square} + \varepsilon$

Where:

- y = target variable (insurance charges)
- $x_1, x_2, \ldots, x_{\square}$ = input features
- β_0 = intercept
- $\beta_1, \beta_2, \ldots, \beta_{\square}$ = regression coefficients
- ε = error term

The model estimates the coefficients that minimize the sum of squared errors between actual and predicted values.

Ridge Regression

Ridge Regression is a regularized version of linear regression that adds an **L2 penalty** to reduce overfitting.

Objective Function:

Minimize: $\Sigma(y_i - \hat{y}_i)^2 + \alpha \Sigma\beta_j^2$

Where:

- α (alpha) is the regularization parameter
- Larger alpha values penalize large coefficients more strongly

Ridge regression helps when features are highly correlated.

Lasso Regression

Lasso Regression introduces **L1 regularization**, which can shrink some coefficients exactly to zero.

Objective Function:

Minimize: $\Sigma(y_i - \hat{y}_i)^2 + \alpha \Sigma|\beta_j|$

This property allows Lasso to perform automatic feature selection and produce simpler models.

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4. Algorithm Limitations

Multiple Linear Regression

- Assumes a linear relationship between variables
- Sensitive to outliers
- Performs poorly when multicollinearity exists

Ridge Regression

- Does not completely remove irrelevant features
- Requires careful selection of alpha value

Lasso Regression

- Can remove important features if alpha is too high
- Unstable when features are strongly correlated

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5. Methodology / Workflow

Step-by-Step Procedure

1. Dataset loading using Pandas
2. Conversion of categorical variables into numerical format
3. Selection of features and target variable
4. Splitting the dataset into training (80%) and testing (20%) sets

5. Training models:
    - Multiple Linear Regression
    - Ridge Regression
    - Lasso Regression
6. Predicting insurance charges on test data
7. Evaluating models using MSE and R^2 score
8. Visualizing predicted vs actual values

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6. Performance Analysis

The models are evaluated using the following metrics:

- **Mean Squared Error (MSE):** Measures the average squared difference between actual and predicted values
- **R^2 Score:** Indicates how well the model explains the variance in the target variable

Observations

- Multiple Linear Regression provides a strong baseline model
- Ridge Regression improves stability by controlling coefficient size
- Lasso Regression simplifies the model by eliminating less important features

All models show a good fit, indicating that factors such as smoking status, BMI, and age significantly influence insurance charges.

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7. Hyperparameter Tuning

Ridge Regression

- Hyperparameter: α (alpha)
- Controls the strength of L2 regularization
- Higher values reduce overfitting but may increase bias

Lasso Regression

- Hyperparameter: α (alpha)
- Controls sparsity of coefficients
- Higher values result in more coefficients becoming zero

Tuning Approach

- Different alpha values were tested
- The value producing the best balance between bias and variance was selected
- Proper tuning improved generalization and reduced overfitting

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CODE:

```
# ================================
# IMPORT LIBRARIES
# ================================
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error, r2_score


# ================================
# LOAD DATASET
# ================================
df = pd.read_csv("medical_insurance.csv")

print(df.head())
print(df.columns)


# ================================
# ENCODE CATEGORICAL DATA
# ================================
# sex, smoker, region are categorical
df = pd.get_dummies(df, drop_first=True)


# ================================
# FEATURES & TARGET
# ================================
X = df.drop("charges", axis=1)   # features
```

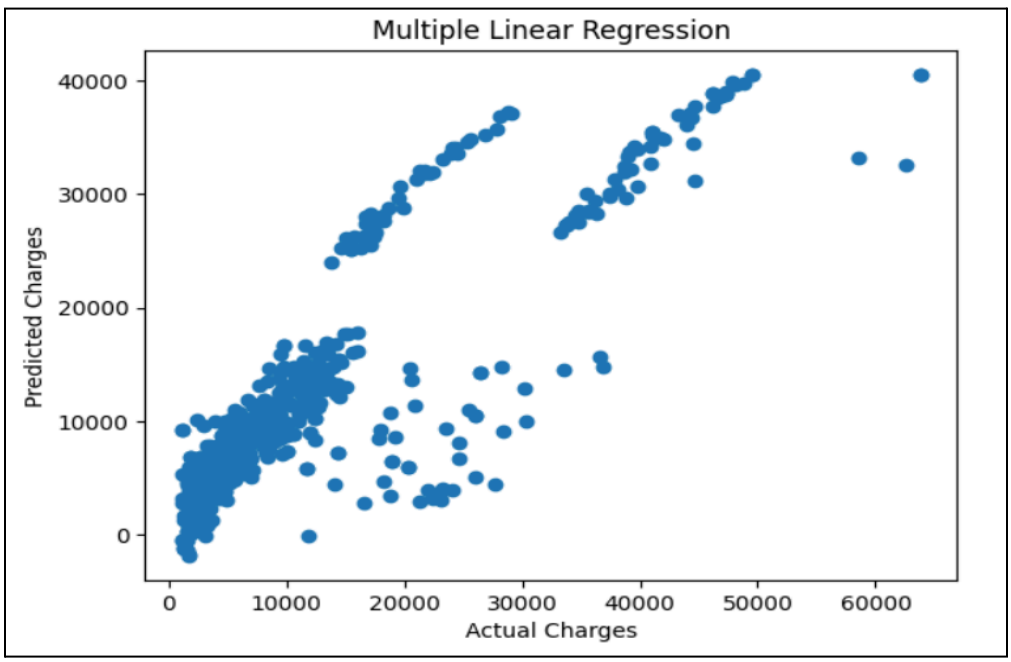
```python
y = df["charges"]                # target


# ================================
# TRAIN TEST SPLIT
# ================================
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


# ================================
# MULTIPLE LINEAR REGRESSION
# ================================
mlr = LinearRegression()
mlr.fit(X_train, y_train)
y_pred_mlr = mlr.predict(X_test)

print("\nMultiple Linear Regression")
print("MSE:", mean_squared_error(y_test, y_pred_mlr))
print("R2:", r2_score(y_test, y_pred_mlr))


# ================================
# RIDGE REGRESSION
# ================================
ridge = Ridge(alpha=1.0)
ridge.fit(X_train, y_train)
y_pred_ridge = ridge.predict(X_test)

print("\nRidge Regression")
print("MSE:", mean_squared_error(y_test, y_pred_ridge))
print("R2:", r2_score(y_test, y_pred_ridge))


# ================================
# LASSO REGRESSION
# ================================
lasso = Lasso(alpha=0.1)
lasso.fit(X_train, y_train)
```

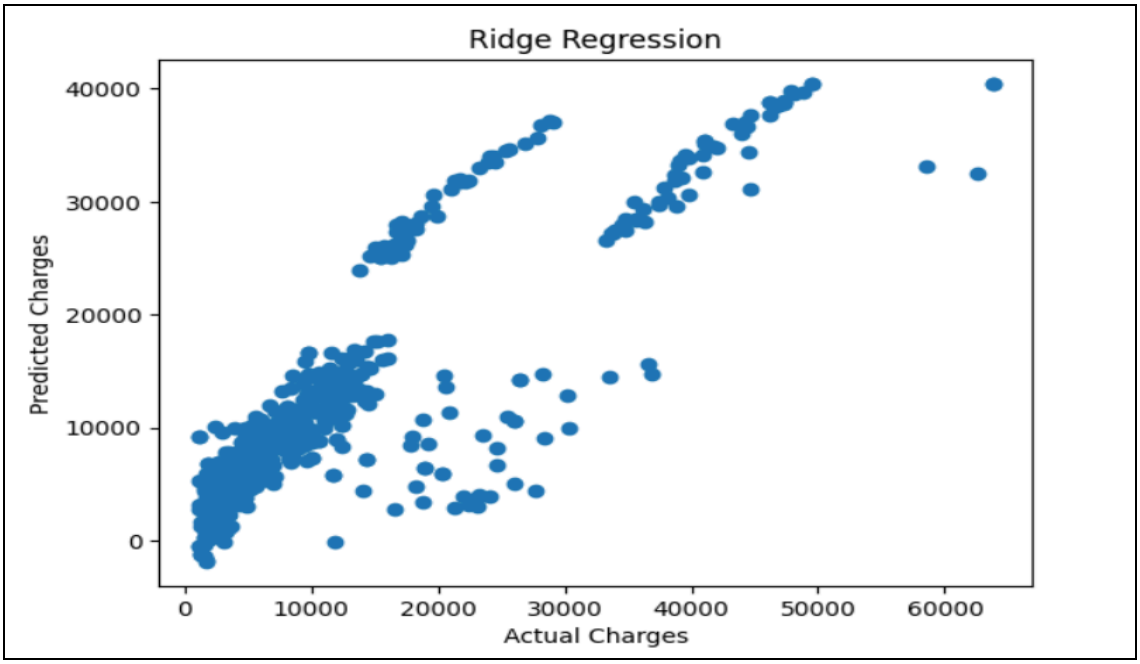
```python
y_pred_lasso = lasso.predict(X_test)

print("\nLasso Regression")
print("MSE:", mean_squared_error(y_test, y_pred_lasso))
print("R2:", r2_score(y_test, y_pred_lasso))


# ================================
# VISUALIZATION
# ================================

# Linear Regression
plt.scatter(y_test, y_pred_mlr)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Multiple Linear Regression")
plt.show()

# Ridge Regression
plt.scatter(y_test, y_pred_ridge)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Ridge Regression")
plt.show()

# Lasso Regression
plt.scatter(y_test, y_pred_lasso)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Lasso Regression")
plt.show()

# Comparison Graph
plt.scatter(y_test, y_pred_mlr, label="Linear")
plt.scatter(y_test, y_pred_ridge, label="Ridge")
plt.scatter(y_test, y_pred_lasso, label="Lasso")

plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Model Comparison")
```

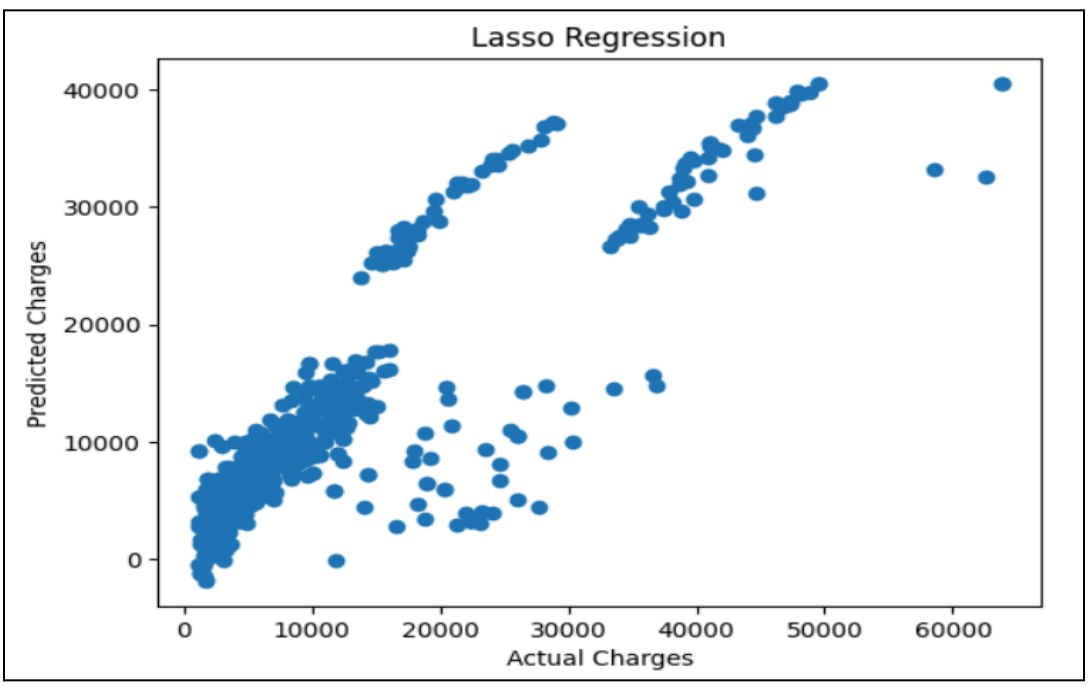
```
plt.legend()
plt.show()
```

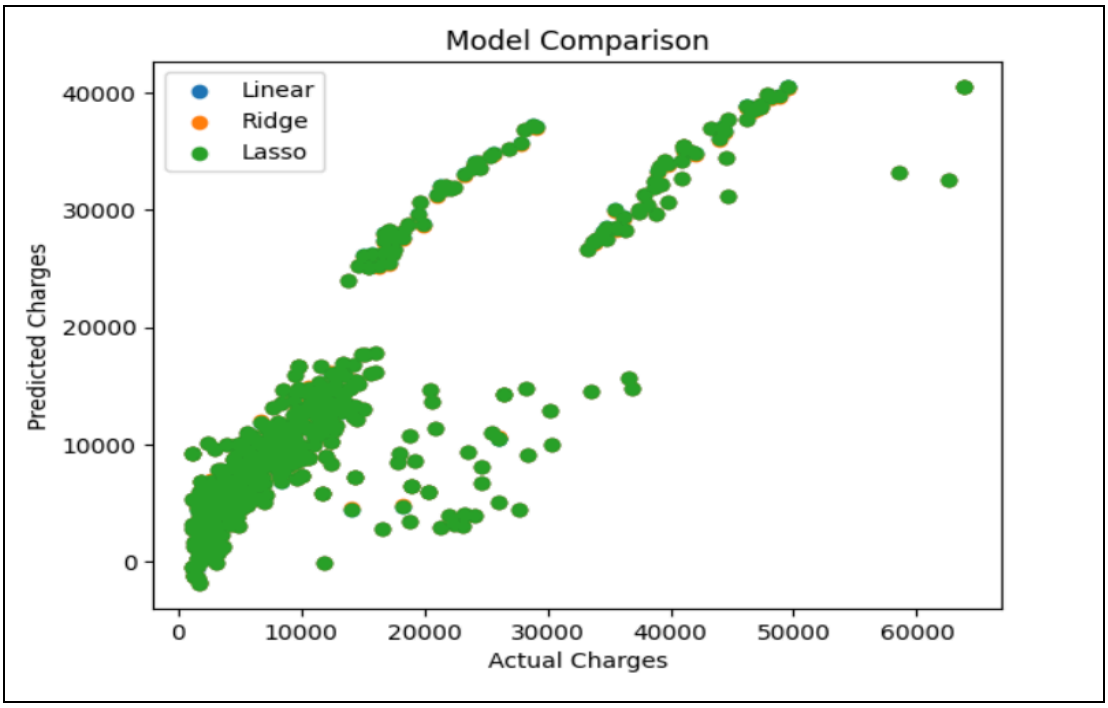
OUTPUT:

```
   age     sex     bmi  children smoker     region      charges
0   19  female  27.900         0    yes  southwest  16884.92400
1   18    male  33.770         1     no  southeast   1725.55230
2   28    male  33.000         3     no  southeast   4449.46200
3   33    male  22.705         0     no  northwest  21984.47061
4   32    male  28.880         0     no  northwest   3866.85520
Index(['age', 'sex', 'bmi', 'children', 'smoker', 'region', 'charges'], dtype='object')

Multiple Linear Regression
MSE: 39933194.54805147
R2: 0.73981661775643

Ridge Regression
MSE: 39943138.4742946
R2: 0.7397518284403735

Lasso Regression
MSE: 39933175.7909889
R2: 0.7398167399674386
```




Ridge Regression
Predicted Charges
Actual Charges
0
10000
20000
30000
40000
50000
60000


Lasso Regression
Predicted Charges
Actual Charges
0
10000
20000
30000
40000
50000
60000



Conclusion

This experiment successfully demonstrated the use of Multiple Linear Regression, Ridge Regression, and Lasso Regression on a real-world medical insurance dataset. Regularization techniques helped improve model robustness and interpretability. The results show that healthcare costs can be effectively predicted using regression models when appropriate preprocessing and tuning techniques are applied.